



I'm a graduate student with a strong background and passion in computer programming, an interest in science & research, and content creation; born in Bratislava (Slovakia), now based in Gothenburg (Sweden). I'm particularly interested in (functional) programming, software verification, linguistics, design, and teaching. I'm also an active contributor (5k+ contributions) of various projects on GitHub. I've been a private tutor in science for several years and have delivered 250+ lectures.

Education

Master of Computer Science, *Chalmers University of Technology* Expected 2027

- Specialization: functional programming, compiler construction, type theory.

Bachelor of Software Engineering, *Chalmers and Gothenburg University* 2022 – 2025

- Honors: Graduated with distinction; minor in Mathematics (60 ETCS)
- Thesis: "Evaluating Trade-offs of Quantized LLMs for Requirements and Test Alignment"*
- Organized Hackathons, workshops and represented students in decision-making bodies through the student union.

Work Experience

Software Engineering Intern Mar 2025 – Sep 2025

Tesla, Inc. *Amsterdam, Netherlands*

- Helped to develop and design diagnostic interfaces for Tesla car touchscreens for vehicle maintenance and repair.[§]
- Developed software instruments that facilitate service automation for (among others) Robotaxis by connecting service & development teams.
- *Technologies used*: TypeScript + React, Rust, Haskell, Shell + Linux

Research & Teaching Assistant Sep 2023 – Mar 2025

Chalmers University of Technology *Gothenburg, Sweden*

- Evaluated the impact of (i) AI-assisted tools, (ii) social annotations for programming in higher education
- Assisted with administrative tasks and coursework assignments and supervised students in courses: Object-Oriented Programming, C & Embedded and Realtime Systems, and Requirements Engineering.
- *Technologies used*: C/C++, Java, R, MATLAB

Developer Oct 2024 – Mar 2024

Algeno AB *Gothenburg, Sweden*

- Focused on developing AI-driven tools to optimize energy use and temperature control in homes and workplaces.
- *Technologies used*: TypeScript + Next.js, Python, pSQL, Docker

IT Consultant & Developer Dec 2022 – Nov 2023

Chalmers Teknologkonsulter AB *Gothenburg, Sweden*

- Led a project for a mobile-first application for the Regional Cancer Center (Sweden) aimed to help detect melanoma among patients.
- *Technologies used*: TypeScript + React Native, Scala

Fullstack Web Developer May 2021 – Jun 2023

Klas, s.r.o. *Bratislava, Slovakia*

- Designed and developed the company's website & trivia platform for the public domain with contemporary visuals.
- *Technologies used*: JavaScript + Next.js, Python, MySQL

Skillset

SCRUM, Agile Workflow, Project Management, DevOps, Robotics, Graphic & Product Design, Teaching, Academic Writing, Languages and Translations.

*To access the preprint: <https://michalspano.com/BSc-thesis>.

§More about the project: <https://service.tesla.com/en-US/service-mode>.

Awards & Certificates

- Winner of *ActInSpace* Hackathon (Sweden), 2022
 - Swedish as Second Language, *Folkuniversitet*, 2024
- Harvard’s CS50 Introduction to Computer Science
 - The Duke of Edinburgh’s Award; TOEFL iBT

Languages: Slovak, English (Native); Czech, Swedish (*C1*); German, Dutch (*B1-B2*)

Projects

michalspano.com/regex	2025	michalspano.com/AppointDent	2024
– An interactive regex parser and an interpreter for a subset of POSIX regex written in Haskell. – The project features a vizualizer of the underlying data structures.		– A microservices-based full-stack webapp to manage dentist appointments. – It provides tools for dentists to organize their schedule and communicate with clients.	
skipgu.org/past-exams	2023 – 2026	michalspano.com/SEManager	2023
– An exam archival system with a dedicated CI/CD pipeline which generates a searchable static website of exam material for students.		– An interactive study planner for Software Engineering students at Gothenburg University. – It features a comprehensive view with all course information and pre-requisites among courses (represented as a graph-like data structure).	